

Fresh critical audit of the Glimm–Jaffe research program

Severity-ranked proof check, source verification, and closure workflow

Audit target: `glimm_jaffe/m6/main.tex`

9 August 2026

Outcome first.

- **Central Main Theorem 9.1: not refuted, but not certification-ready as written.** After the explicit local repairs recorded in this report and proved in the two companion Lamport documents, its mathematical dependency chain closes conditionally on the enumerated cutoff-construction and GJS inputs.
 - **Corollary 9.3 (“no symmetry breaking”): not proved as stated.** Its only cited internal theorem assumes a phase-decomposition hypothesis that Corollary 9.3 omits. This is the audit’s one confirmed proof-breaking issue. It does not feed back into Main Theorem 9.1.
 - **Citation correctness: three load-bearing pinpoint defects.** Both FKN citations to Simon Theorem III.12 are wrong; the hypercontractivity theorem numbers are wrong; and the cited domain-smoothing theorem has a hypothesis not verified in the manuscript. Correct source theorems exist and repair all three imports.
 - **Local derivation defects: repairable.** The manuscript has a false unit-cell count, a false inequality for negative Euclidean time, an incomplete positivity choice in the moment transfer, a dropped prefactor, an unrestricted spectral parameter, and omitted signed truncation details. None changes the qualitative theorem after correction.
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1 Frozen audit object and standard of review

The canonical target was fixed before the audit:

```
glimm_jaffe/m6/main.tex,
SHA256 = fdd67e696e68f93f5227ddfe32a6927a
ace0099dc969819417ddcc9eca3f886a.
```

The source has 1,022 lines. A fresh `latexmk -pdf` build produced a 15-page PDF with no unresolved references; layout warnings were limited to overfull boxes. Successful typesetting is not treated as mathematical verification.

The review standard was:

- R1. Every conclusion must be reachable from earlier displayed assumptions by an explicit logical or analytic step.
- R2. Every use of an unbounded operator must state the relevant domain, closure, or truncation passage.
- R3. Every limit must identify its topology and the theorem permitting the passage.
- R4. Every source citation must point to a result whose statement has the hypotheses and conclusion actually used.
- R5. Contextual literature claims must not be promoted to mathematical inputs.
- R6. A defect is called *proof-breaking* only if the claim does not follow from the stated hypotheses. A defect with a proved local correction is called *repairable*.

2 Agentic workflow and gates

The work was executed as a stateful audit workflow. Each state had a testable exit gate; failure returned the object to the preceding state rather than being concealed by prose.

State	Action	Concrete output	Exit gate
A0	Freeze target	Canonical source, line count, SHA-256, fresh build	One unambiguous proof version
A1	Reconstruct dependencies	Theorem-to-lemma graph and import ledger	Every main conclusion has predecessors
A2	Local adversarial check	Recomputed constants, domains, signs, limits, spectral supports	Every internal arrow either passes or has a counterexample/repair

State	Action	Concrete output	Exit gate
A3	Source acquisition	All 15 bibliography entries present locally; three missing modern/method references downloaded afresh	Every citation has a local source artifact
A4	Pinpoint source check	Exact theorem pages and scope comparison	Used statement is no stronger than source statement
A5	Proof-break search	M3 hypothesis trace; historical and uniqueness scope audit	Unstated hypotheses isolated
A6	Corrected reconstruction	Two independent Lamport-structured proofs	Every repair integrated and cross-referenced
A7	Artifact QA	Compile, render every page, inspect every rendered page, clean temporary files	No TeX errors, clipping, blank pages, or unreadable evidence

3 Dependency graph and audit result

The manuscript’s central proof has the following actual dependency graph:

$$\begin{aligned}
&\text{cutoff self-adjointness, simple positive ground state, FKN, domains} \implies (M_2) \text{ uniform local moment} \\
&\quad + \text{ GJS CE1, CE2, cutoff uniformity, scaling} \implies (G) \text{ uniform cutoff gap} \\
\\
&(M_2) + (G) + \text{finite propagation + patching} \implies \text{LPPL estimate} \implies \text{full-net local-norm convergence} \\
&\quad + \text{affiliation + path regularity} \implies \text{local normality} \\
\\
&\text{local-norm convergence + patching + covariance} \implies \text{spectral ground state} \\
&\quad \text{translation and parity invariance.}
\end{aligned}$$

The two GJS estimates are the only *nonstandard weak-coupling estimates* in this graph. They are not the only logical inputs. Therefore line 895’s statement that all conclusions are “conditional only” on GJS Theorems 1.1.7–1.1.8 is false as a literal dependency claim.

Claim	Verdict	Reason
Uniform local moment (M_2)	REPAIR	Correct after choosing $F_1 = F_2 = V(h) $ and writing the unbounded truncation/weak-graph passage.
Uniform gap (G)	REPAIR	CE2-to-semigroup argument and density argument pass after the cell-count, time-sign, moment-prefactor, and β repairs.
Affine-path differentiability	REPAIR	Argument passes with correct FKN/hypercontractivity citations and model-specific domain theorem.
Exact spectral filter	PASS	Fourier sign, inverse-energy multiplier, integrability near 0, and rapid tail all check.
Unbounded clustering	PASS	Spectral supports, locality identity, affiliation, and Gaussian separator constants check.
One-path LPPL estimate	PASS	Factor $32M$, distance loss, and shell count recompute correctly.
Full-net Cauchy limit/rate	PASS	Cofinality and local dual completeness are used correctly.
Local normality	PASS	Direct monotone-net argument proves norm-closedness in the required case.
Spectral ground-state passage	PASS	The exact finite-volume spectral identity and the two-limit argument check.
Translation/ $\mathbb{Z}_2$ invariance	PASS	Covariance, cofinality, simplicity, and local norm passage check.

Canonical averaged vacuum	PASS	The shifted plateaux contain $[-2n, 2n]$ uniformly in the averaging parameter.
No symmetry breaking, Cor. 9.3	FAIL	Companion M3 Theorem 4.2 assumes a broken-phase structure hypothesis absent from the corollary.

4 Confirmed proof-breaking issue

4.1 Corollary 9.3 omits the hypothesis used by M3

The formal inference in the manuscript is invalid.

⟨1⟩ 1 What Corollary 9.3 states.

For an even polynomial in the weak-coupling region, it says the uniform gap is incompatible with coexistence of two clustering phases exchanged by parity, citing M3, Theorem 4.2.

⟨1⟩ 2 What M3, Theorem 4.2 assumes.

Lines 239–244 of `m3/m3.tex` impose Hypothesis 4.1: two translation-invariant clustering ground states $\omega_{\pm}$, parity exchange, a nonzero local order parameter, and—critically—identification of every parity-invariant ground state under discussion with the symmetric midpoint $\frac{1}{2}(\omega_{+} + \omega_{-})$. The theorem begins “Assume P even and Hypothesis 4.1.”

⟨1⟩ 3 Where the missing hypothesis is used.

M3 first obtains a parity-invariant clustering limit ω . It then invokes the hypothesis to write

$$\omega = \frac{1}{2}(\omega_{+} + \omega_{-}).$$

For an order parameter with $\omega_{\pm}(A) = \pm m$, this gives

$$\omega(A) = 0, \quad \lim_{|a| \rightarrow \infty} \omega(A\tau_a(A)) = m^2 \neq 0,$$

contradicting clustering. Without the midpoint identity, neither displayed equality follows merely from parity invariance.

⟨1⟩ 4 Logical diagnosis.

M3 proves

$$\text{evenness} + \text{Hypothesis 4.1} + \text{uniform gap} \implies \perp.$$

Corollary 9.3 attempts to use this as

$$\text{evenness} + \text{uniform gap} + \text{two exchanged clustering states} \implies \perp,$$

which drops a premise. This is the invalid inference $(A \wedge B \Rightarrow C) \not\Rightarrow (A \Rightarrow C)$.

⟨1⟩ 5 Exact repair choices.

Either append all of M3 Hypothesis 4.1 to Corollary 9.3, or delete the corollary and retain the proved parity invariance of the constructed cutoff-net limit. The GJS 1975–1976 phase-transition papers do not, as cited in M6, automatically supply the exact Hamiltonian phase-space classification needed here.

5 Load-bearing source mismatches

5.1 FKN: Simon Theorem III.12 is the wrong theorem

M6 lines 305 and 536 cite Simon Theorem III.12 for FKN. The page image shows that Theorem III.12 is an estimate for products of projections attached to Sobolev/Markov regions. It does not state either slab identity or the interacting semigroup formula.

The correct source chain is:

Simon Theorem III.6	free-field FKN formula,
Simon Corollary V.14 and Theorem V.15	interacting FKN formula,
GRS Theorems II.14 and II.16	cutoff and general FKN formulas.

The mathematical formula used in M6 is supported by these results, so this is a repairable citation failure, not a refutation of the argument.

5.2 Hypercontractivity: the exact theorem is Simon I.17/III.17

M6 line 542 cites Simon Theorems V.10 and V.12 for

$$\|P_T F\|_{p'} \leq \|F\|_p, \quad \frac{p' - 1}{p - 1} = e^{2m_0 T}.$$

Simon Theorem I.17 is the free Nelson hypercontractivity theorem, and Theorem III.17 is its time-region form. Theorem V.10 uses the earlier estimate inside a transfer-matrix argument; Theorem V.12 is an essential-self-adjointness theorem. The exponent arithmetic in M6 is correct, but the pinpoint citation is not.

5.3 Domain smoothing: use Expositions Theorem 7.4

M6 line 214 cites Theorem 7.3 and Lemma 7.4 to infer $\text{Ran } e^{-tK(g)} \subset D(V(h))$. Theorem 7.3 assumes the multiplication potential V itself is bounded below. M6 does not establish that assumption for the cutoff Wick interaction. The next result, Expositions Theorem 7.4, is explicitly model-specific and states

$$\text{Ran } e^{-tH(g)} \subset D(H_0) \cap D(V')$$

for every relevant polynomial multiplication V' . Citing Theorem 7.4 repairs the domain input exactly.

6 Repairable internal defects with explicit derivations

6.1 False unit-cell bound

M6 lines 376–377 use $L + 1$ as an upper bound for the number of unit cells met by an interval of length L . Let $I = (-0.1, 0.1)$; it has length 0.2 and meets both $[-1, 0)$ and $[0, 1)$ in positive measure, while $L + 1 = 1.2 < 2$. Thus the inequality is false.

For half-open cells $[j, j + 1)$, the correct integer bound is

$$N_I \leq \lfloor L \rfloor + 2 \leq L + 2. \quad (6.1)$$

Consequently the two displayed estimates must read

$$\begin{aligned} \|Q\|_{\text{GJS}} &\leq K_1 K_2^{2r} r! (L + 2)^{r/2} \prod_i \|f_i\|_2, \\ \|Q^2\|_{\text{GJS}} &\leq K_1 K_2^{4r} (2r)! (L + 2)^r \prod_i \|f_i\|_2^2. \end{aligned}$$

Only vector-dependent constants change; the decay exponent m_1 and gap proof do not.

6.2 Wrong time-sign inequality

M6 line 412 bounds $\|\psi_{T+s}\|$ by $\|\psi_T\|$ for both $s = \pm t/2$. Since $a \mapsto \|e^{-aH} \Omega_0\|$ decreases, this holds for $s \geq 0$ and reverses for $s < 0$.

The corrected bound is

$$\frac{\|\psi_{T+s}\|}{\|\psi_T\|} \leq \begin{cases} 1, & s \geq 0, \\ c_g^{-1}, & s < 0, \end{cases} \quad c_g = \langle \Omega_g, \Omega_0 \rangle > 0. \quad (6.2)$$

For fixed g and fixed s , both ratios tend to 1 as $T \rightarrow \infty$. The finite bound (6.2), not uniformity in g , is all that the weak-graph passage needs.

6.3 Nonnegative FKN arguments at $t = 0$

The slab lemma is stated for individually nonnegative F_1, F_2 . M6 line 352 specifies only their product. Set

$$F_1 = F_2 = |V(h)|.$$

Then both hypotheses hold and $F_1 F_2 = V(h)^2$. Bounded truncations $|V(h)| \wedge M$ give the identity first; monotone convergence and CE1 give the unbounded limit.

6.4 Signed truncations in the gap transfer

For a real polynomial multiplication operator Q , define

$$Q^{[M]} = (-M) \vee (Q \wedge M).$$

Write $Q^{[M]} = (Q^{[M]})^+ - (Q^{[M]})^-$ and apply the nonnegative FKN identity bilinearly. The two-time product is dominated by

$$|Q(-t/2)Q(t/2)| \leq \frac{1}{2}(Q(-t/2)^2 + Q(t/2)^2). \quad (6.3)$$

CE1 bounds both terms. On Hilbert space, $Q^{[M]}\psi_a \rightarrow Q\psi_a$ follows from $|Q - Q^{[M]}|^2|\psi_a|^2 \leq Q^2|\psi_a|^2$. These statements supply the omitted limit theorem in lines 404–418.

6.5 Dropped K_1 and spectral range

From the corrected moment bound

$$\mathbb{E}_{\mu_g} X^{2n} \leq C_8 K_1 (K')^{2n} (2n)!,$$

one obtains

$$\mathbb{E}_{\mu_g} e^{s|X|} \leq 2 \max\{1, C_8 K_1\} \sum_{n \geq 0} (sK')^{2n}, \quad (6.4)$$

not the expression at line 455 with K_1 absent. The radius $s < K'^{-1}$ is unchanged.

In the final projection argument one must fix $0 \leq \beta < m_1$. For $\beta < 0$, $P_{[0,\beta]}\Omega_g = 0$, contrary to line 487. Restricting β gives

$$e^{-\beta t} \|P_{[0,\beta]}\psi_Q\|^2 \leq \langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq C_Q e^{-m_1 t},$$

and the intended conclusion follows.

6.6 Dimensionless weak-coupling reduction

GJS page 590 supplies the dilation

$$(m_0, \lambda) \mapsto (s^{-1}m_0, s^{-2}\lambda).$$

If the printed expansion is valid at an admissible M_* for $\lambda' < \lambda_*$, choose $s = m_0/M_*$. Then

$$m'_0 = M_*, \quad \lambda' = \lambda M_*^2 / m_0^2.$$

Thus $\lambda/m_0^2 < \lambda_*/M_*^2$ implies the printed hypotheses. This explicit reduction should accompany the definition $\varepsilon_{\mathcal{D}} = \lambda_*/M_*^2$.

7 Source-by-source cross-check

All bibliography entries in M6 are present under `glimm_jaffe/refs`; BDW, RZT, and Lamport were downloaded afresh from their official author/arXiv locations. The separate dossier contains the exact page images described below.

Source	Claim checked	Exact page/result inspected	Verdict
GJ I (1968)	cutoff self-adjointness; locality; cutoff independence in a light cone	article PDF pp. 5–6, especially locality/Trotter discussion	PASS , broad citation
GJ II (1970)	ground state exists and is unique; locality/covariance	Thms. 2.2.1, 2.3.1; printed pp. 397–400	PASS
GJ III (1970)	averaged w^* limit point and locally Fock/normal limit points	printed pp. 210–211, Thms. 2.1–2.3	PASS
GJ IV (1972)	completion of construction/Wightman functions	first-page Theorems A–B	Contextual pass
GJ (1971)	positivity/lower bounds/self-adjointness	Theorems 1–2, PDF pp. 2, 4	Corroborating; not sufficient for every clause alone
Rosen (1970)	self-adjoint cutoff construction	Theorem 5.3 and following core argument, PDF pp. 22–23	Corroborating
GJ Expositions	limit not proved	PDF pp. 54–56 (printed p. 50 and surrounding construction)	PASS
GJ Expositions	finite propagation/patching	same chapter, PDF pp. 55–56	PASS , chapter citation broad
GJ Expositions	semigroup domain smoothing	Theorems 7.3–7.4, PDF pp. 165–167	REPAIR : cite Thm. 7.4
GJS (1974)	cutoff class and scaling	printed pp. 588–590	PASS
GJS (1974)	norm, degree class, CE2, CE1	printed pp. 593–595, (1.1.5)–(1.1.10), Thms. 1.1.7–1.1.8	PASS
GJS (1974)	cutoff uniformity	printed p. 629	PASS for Thm. 1.1.8; CE2 already says independent of h
GRS (1975)	FKN	Thms. II.14 and II.16, Part I PDF pp. 39–40	PASS ; use pinpoint citation
GRS (1975)	Problems 1–2	printed pp. 125–126 and note added p. 128	PASS with historical qualification
Simon (1974)	free/interacting FKN	Thm. III.6; Cor. V.14; Thm. V.15	REPAIR citation
Simon (1974)	purported III.12 FKN citation	Thm. III.12, PDF p. 119	FAIL as citation
Simon (1974)	hypercontractivity	Thm. I.17 and III.17	REPAIR citation
Simon (1974)	exponential stability; common core	Thms. V.7 and V.12	PASS in their actual roles
AHZ (1989)	Euclidean uniqueness/global Markov property in polynomial models	abstract and uniqueness conclusion near PDF p. 24	Contextual pass; hypotheses matter
BDW (2025)	modern uniqueness	abstract: uniqueness of invariant measure for stochastic quantization above critical temperature	Scope mismatch if used for Hamiltonian uniqueness; M6 uses context only

Source	Claim checked	Exact page/result inspected	Verdict
RZT (2025)	Hamiltonian-renormalisation scope	abstract explicitly says finite volume	PASS
GJS (1975–76)	phase transition/two pure phases	opening statements and theorem pages	Contextual pass; does not by itself supply M3 Hypothesis 4.1 as used
Lamport (2012)	hierarchical proof notation	worked examples and numbering discussion	Method followed in companion documents

8 Historical and scope corrections

8.1 GRS Problems 1–2

GRS printed pages 125–126 formulate Euclidean local L^p bounds and convergence of finite-volume Gibbs states. A note added on printed page 128 states that small-coupling $p = 1$ versions were solved by Newman. M6 is justified in distinguishing its Hamiltonian local-von-Neumann-algebra norm convergence from those Euclidean problems, but it should not suggest without qualification that every formulation of Problems 1–2 remained open.

8.2 Modern uniqueness citations

BDW proves uniqueness of an invariant measure for the Φ_2^4 stochastic quantization equation above the critical temperature. It does not itself prove uniqueness of locally normal Hamiltonian ground states or convergence of the specific cutoff Hamiltonian net. AHZ proves Euclidean Gibbs uniqueness/global Markov statements under its stated conditions. M6 appropriately treats these as context, not inputs, but the wording should remain explicit.

8.3 Exhaustive literature claim

The sentence that Hamiltonian local-norm convergence “has remained open in all regimes” is an exhaustive claim about the literature. The finite list cited in M6 establishes that the original construction used limit points and that RZT is finite volume; it does not establish the absence of every later result in every regime. Replace it by the defensible statement: “The cited primary construction does not prove local-norm convergence of this cutoff net, and no such theorem is identified in the references reviewed here.”

9 Severity register

ID	Severity	Location	Finding and mathematical consequence	Disposition
P1	proof-breaking	Cor. 9.3; M3 lines 239–268	Omitted phase-decomposition premise; corollary does not follow.	Add premise or delete
S1	high	lines 305, 536	Simon III.12 is not FKN.	Cite III.6, V.14–V.15 or GRS
S2	high	line 542	V.10/V.12 are not the direct hypercontractivity theorem.	II.14/II.16 Cite I.17/III.17

ID	Severity	Location	Finding and mathematical consequence	Disposition
S3	high	line 214	Theorem 7.3’s bounded-below hypothesis is not checked.	Cite model-specific Thm. 7.4
L1	moderate	lines 376–377	$L + 1$ cell count is false.	Replace by $L + 2$
L2	moderate	line 412	Semigroup norm inequality fails for negative s .	Use sign split and c_g^{-1}
L3	moderate	line 352	Nonnegative inputs not individually specified.	Set both equal to $ V(h) $
L4	moderate	lines 404–418	Signed/unbounded FKN limit under-described.	Use clipped truncations and (6.3)
L5	low	line 455	K_1 dropped from prefactor.	Restore it outside series
L6	low	lines 484–489	β must be nonnegative.	State $0 \leq \beta < m_1$
C1	moderate	standing weak coupling	Dimensional reduction omitted.	Insert exact GJS scaling
C2	moderate	line 895	“conditional only” hides independent imports.	Enumerate all imports
H1	moderate	introduction	“open in all regimes” not established by cited corpus.	Narrow claim
H2	low	GRS discussion	Later $p=1$ note not mentioned.	Add qualification

No P0 defect—a defect that survives all local repairs and breaks Main Theorem 9.1—was found. This negative finding is limited to the frozen source and imported theorems checked; it is not a substitute for independent journal peer review.

10 Closure plan for the research program

The remaining work is finite and ordered by logical risk.

- C1. **Patch the theorem-bearing source.** Apply S1–S3, L1–L6, and C1–C2 exactly as written in the two Lamport proofs. Acceptance test: a line-by-line comparison finds no use of the old constants, time-sign inequality, or theorem numbers.
- C2. **Resolve Corollary 9.3.** Either state all of M3 Hypothesis 4.1 or remove the corollary. Acceptance test: every premise used in the midpoint contradiction appears in the corollary statement.
- C3. **Narrow the literature claims.** Incorporate the GRS note and replace the exhaustive modern-open-problem assertion. Acceptance test: every historical sentence is entailed by a displayed primary-source page.
- C4. **Re-run the structured proof.** Use the two companion Lamport documents as a checklist against the patched M6 source. Acceptance test: each $\langle 1 \rangle$ step maps to a unique lemma/proposition in M6.
- C5. **Independent source audit of the only irreducible analytic import.** Have a second reader verify GJS (1.1.7)–(1.1.10), especially the localization-square convention at sharp times and the cutoff-uniform scope. Acceptance test: signed page references and recomputed constants agree with the dossier.
- C6. **Certification build.** Compile, run a reference/warning check, render every page, and inspect every page. Acceptance test: no unresolved references, clipped equations, or unexplained external hypotheses.

11 Reference artifact manifest

The full filenames are retained in `glimm_jaffe/refs`. The following SHA-256 values make the source audit reproducible; long titles are abbreviated only in this table.

Artifact	SHA-256
GJ I (1968)	80993a0521b7f0de21b096c93365b3d8 4cd569ed457dcb71b5766866b9c8ac07
GJ II (1970)	6c33fbc138766c344ee9eaca8bd78089 3108a74549c8de74d995f08ec75f05d1
GJ III (1970)	8bfa327da9f3a78222243b3990b44693 bd8e2cf1737c7a7d6f8749cfd3d9f961
GJ IV (1972)	b073f26cc15e08e66b4ea8480489109d 7537f00e83ebfffd457cb62151483774
GJ positivity (1971)	655df0cadb54e132594d85a27b184971 9128e8604fb4161f418a8d56d156ab97
GJ Expositions	d474345dc9336e968728e6ca2620d05f 41ad21deb85143b48cceb34b21410fe
GJS (1974)	e0a08c9438ff5c69f6d55f1c1b48e02 01fc44f2313cef94ad4e87deaf872aa48
GJS phase transitions (1975)	dc20756fcdfa18992bb1034a35415d008 e02339718beeca06d0ab74ec71c40e9
GJS mean field I (1976)	ef7957c5ef3eef3d867cf8db676bc9cb f03944d2f6ba79df3c68ca293fdc7b2e
GJS mean field II (1976)	7ffaff7052ea5f4dd1826fb1991f4eb6 db9347275f150241ac6603cad2e47d64
GRS Part I (1975)	a97248302fe44eb9cddd690505c319f1 bc9237e87f188fbefd74e4564e9945c5
Rosen (1970)	7336ed96d62107cd10459bd065199c06 76c8b417fc9e74c450a20488f6c953ca
Simon monograph	06dc3239715cebb6a52ceec707e42bcc d0561ed2601370568d29790c3c59133c
AHZ (1989)	a4970d07869341427dd201a9fa397b8c 8be825bb880234dfce61929882452165
BDW arXiv:2504.08606	e1b9ebe4a8a702153e764b3b4a3e9b04 e2adc9818725d0ebb3afeaf3cc954823
RZT arXiv:2505.13030	50eaa5c031bcce2cf3ea057e2d6a9c3 c793fe2734b4b88cae013d02fbf7cec23
Lamport proof article	c0fceb68ecfb1117138c39a942d61e3f 17fb2f0c825067e18c9c657c393049a1

12 Final audit verdict

The highest-value correction is to withdraw or conditionalize Corollary 9.3. The next highest-value work is citation and domain repair, because the present pinpoint references do not certify the formulas they are attached to. Once those changes and the six local derivation corrections are made, the central proof program has a complete conditional route from the printed GJS estimates to the claimed cutoff-net limit. The word “conditional” remains essential: the proof imports the cutoff construction, FKN, semigroup smoothing, finite propagation, patching, covariance, and GJS itself.